



Minimum Not Divisible (notdivisible)

You have a tree (a connected acyclic graph) of N nodes indexed from 1 to N . Each node i has an integer value $V_i > 1$ assigned to it.

You are given Q queries, where each query i consists of two integers A_i and B_i . Your task is to find the minimum value (*greater than 1*) that is not divisible by any of the assigned values on the path between A_i and B_i , including the values assigned to A_i and B_i .

🔗 Among the attachments of this task you may find a template file `notdivisible.*` with a sample incomplete implementation.

Input

The first line contains one integer N , the number of nodes. The second line contains N integers $V_1, V_2, \dots, V_N$.

Each of the following $N - 1$ lines contains two integers a and b , meaning that there is an edge between a and b .

The next line contains one integer Q , the number of queries. Each of the next Q lines contains two integers A_i and B_i , describing a query.

Output

For each query, you need to write a single line with an integer representing the answer to that query.

Constraints

- $2 \leq N \leq 100\,000$.
- $2 \leq V_i \leq 1\,000\,000$ for each $i = 1 \dots N$.
- $1 \leq Q \leq 100\,000$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.
- **Subtask 2** (15 points) $N, Q \leq 1000$.
- **Subtask 3** (20 points) $V_i \leq 3$ for each $i = 1 \dots N$.
- **Subtask 4** (15 points) All V_i values are unique for $i = 1 \dots N$.
- **Subtask 5** (20 points) Every node is connected to at most 2 other nodes.

Examples

input	output
9	5
7 25 8 4 1000000 6 11 3 2	2
5 7	3
5 1	
5 6	
7 3	
1 2	
1 4	
6 8	
2 9	
3	
8 9	
3 8	
4 9	

Explanation

- In the **sample case**:
- Node values on the path of the first query are 3, 6, 1000000, 7, 25, 2. The minimum integer not divisible by any of them is 5.
 - Node values on the path of the second query are 8, 11, 1000000, 6, 3. The minimum integer not divisible by any of them is 2.
 - Node values on the path of the third query are 4, 7, 25, 2. The minimum integer not divisible by any of them is 3.

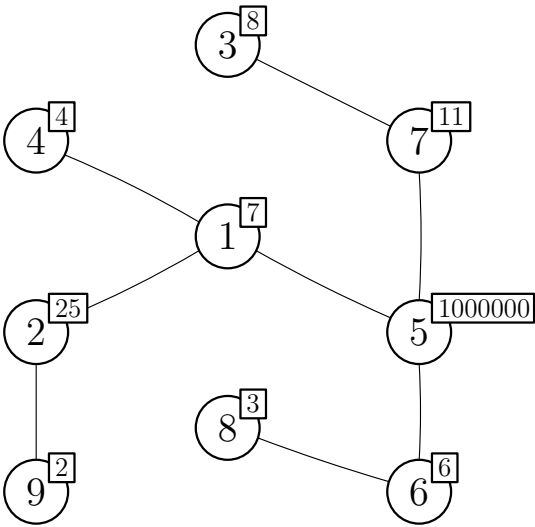


Figure 1: The sample case.